Waterbird Counter

Count mallards, coots and Egyptian geese from a single photo

SDG 15: Life on Land

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Wildlife is declining, and freshwater fastest

73%

average decline of monitored wildlife populations, 1970 to 2020

85%

decline in freshwater populations, the steepest of all habitats

35%

decline in Europe and Central Asia, so this is our problem too

Population declines are an early warning for ecosystems. But you can only see a decline if you count reliably, again and again.

Source: WWF Living Planet Report 2024 (Living Planet Index, ZSL)

Counting waterbirds is still done by eye

Our user

Volunteer waterbird counters, like the thousands who count for Sovon Vogelonderzoek Nederland. Their counts feed national bird trends.

~250,000 mallard pairs in NL

~22,000 Egyptian goose pairs, exotic

Big flocks

Counting 40 birds on moving water by eye is slow and error-prone.

Double counting

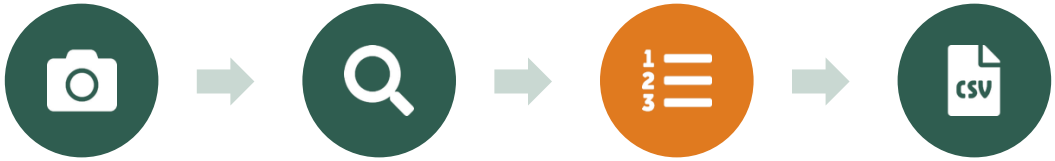
Birds swim, dive and fly between spots during the count.

Exotic species

Egyptian goose first bred near Den Haag in 1967 and is now everywhere. It needs watching.

Sources: Sovon, Verschenen of verdwenen; De Levende Natuur

One photo in, a count per species out



Upload
a photo of a pond or canal

Detect
a box around every bird

Count
per species, exotic geese flagged

Log
per location, export as CSV

From counting by eye to counting in seconds, with the volunteer checking the result.

Live demo

Streamlit app on our Roboflow model

Built on our own footage from Dutch ponds

Film — 4 videos at city ponds, 80 seconds total

Label — ~320 frames + ~40 scraped photos, 3 classes by hand

Train — Roboflow RF-DETR Small, split 70/20/10

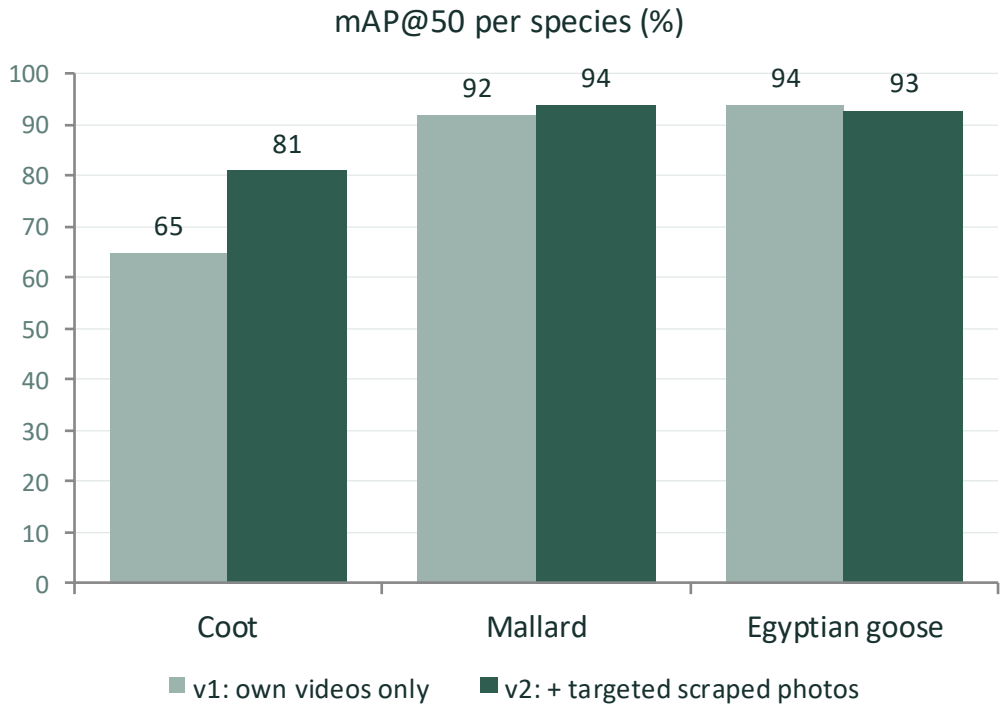
Wrap — Streamlit app calls the model via the API

3 species: mallard, coot, Egyptian goose

~360 labelled images

90.8% mAP@50 on the test set

90% on paper, wrong on new photos



The problem

On internet photos, coots were called mallards. Frames from the same 4 videos sat in training and test data, so the score was too optimistic.

The fix

We added ~40 targeted photos of coots and mallards and retrained.

The result

Coot went from 65% to 81%, and the internet photos were now classified right.

The model counts, the human decides

False trends

Undercounting coots could make a stable population look like it is declining, and send money to the wrong place.

Wrong target

A false Egyptian goose detection could support control measures against birds that were never there.

People in photos

Park photos can show people, so the app stores counts only, never images.

Limitations are shown in the app. The volunteer checks every count before reporting it, so faster counts become reliable data for SDG 15.